

Haptic Wristband

Builder Assembly Guide — XIAO ESP32C3 + Dual Motor

Overview

This guide covers assembly of a WiFi-triggered haptic wristband. The wristband receives UDP packets over WiFi and fires two independently controlled vibration motors. No coding is required — firmware binary is provided. Your job is to wire the circuit, flash the firmware, test it, and mount everything to the wristband.

Parts List (all supplied)

Seeed Studio XIAO ESP32C3 (Pre-Soldered)	1	Main microcontroller
Vibration motor (coin type)	2	One per wrist position
2N2222 NPN transistor	2	One per motor
1N4001 flyback diode	2	One per motor — CRITICAL
220 Ohm resistor	2	Base resistor for each transistor
330 Ohm resistor	1	Status LED current limiter
Status LED	1	Boot/WiFi indicator
JST PH 2.0 female connector	1	Solder to XIAO BAT+/BAT- pads
3.7V LiPo battery w/ JST PH 2.0	1	2000mAh — check polarity before connecting
TP4056 USB charging board	1	Between battery and XIAO power
Elastic wristband	1	Final mounting
Foam padding	1	Between motors and skin
Heat shrink tubing		neededCover all solder joints
Jumper wires		neededPrototyping connections

Note: The XIAO ESP32C3 does NOT have a JST battery socket pre-installed. You must solder the JST PH 2.0 female connector to the BAT+ and BAT- pads on the underside of the board. Check polarity with a multimeter before connecting the battery.

Wiring Reference

Each motor has an identical driver circuit. Motor A uses GPIO D0 (GPIO2). Motor B uses GPIO D10 (GPIO10). Wire both circuits identically, only the GPIO pin differs.

Motor A Circuit (GPIO D0 / GPIO2)

XIAO GPIO D0	220 Ohm resistor leg 1	Control signal
220 Ohm resistor leg 2	2N2222 Base (pin 2 / middle)	Current-limited gate

2N2222 Emitter (pin 1 / left)	XIAO GND	Transistor return
2N2222 Collector (pin 3 / right)	Motor A (negative wire)	Switched ground
Motor A (positive wire)	XIAO 5V / VIN rail	Motor power
1N4001 Cathode (stripe end)	Motor A positive terminal	Flyback protection
1N4001 Anode	Motor A negative terminal	Flyback protection

Motor B Circuit (GPIO D10 / GPIO10)

XIAO GPIO D10	220 Ohm resistor leg 1	Control signal
220 Ohm resistor leg 2	2N2222 Base (pin 2 / middle)	Current-limited gate
2N2222 Emitter (pin 1 / left)	XIAO GND	Transistor return
2N2222 Collector (pin 3 / right)	Motor B (negative wire)	Switched ground
Motor B (positive wire)	XIAO 5V / VIN rail	Motor power
1N4001 Cathode (stripe end)	Motor B positive terminal	Flyback protection
1N4001 Anode	Motor B negative terminal	Flyback protection

Status LED Circuit

XIAO GPIO D6	330 Ohm resistor leg 1	LED control signal
330 Ohm resistor leg 2	LED anode (+)	Current-limited
LED cathode (-)	XIAO GND	LED return

Power Wiring

--	--	--

LiPo battery JST	JST female on XIAO underside	Check polarity first — reversed
TP4056 OUT+	XIAO 5V pin	Optional: use TP4056 for manag
TP4056 OUT-	XIAO GND	Common ground

2N2222 Transistor Pin Reference

Hold the transistor with the flat face toward you, pins pointing down. Left to right:

Pin 1 — left	Emitter	GND
Pin 2 — middle	Base	220 Ohm resistor from GPIO
Pin 3 — right	Collector	Motor negative wire

1N4001 Diode Orientation — CHECK BEFORE EVERY POWER-UP

The stripe on the 1N4001 marks the CATHODE. Cathode (stripe) connects to motor positive terminal. Anode (no stripe) connects to motor negative terminal. A reversed diode will pass current and may damage components immediately. If the diode gets warm on power-up — cut power, re-check orientation.

Firmware Flashing Instructions

Step 1 — Install Arduino IDE

Download Arduino IDE 2.x from: <https://www.arduino.cc/en/software>

Step 2 — Add ESP32C3 Board Support

Open Arduino IDE. Go to File > Preferences. In 'Additional boards manager URLs' add:

```
https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
```

Then go to Tools > Board > Boards Manager, search 'esp32', install the Espressif package.

Step 3 — Select the Board

Tools > Board > ESP32 Arduino > XIAO_ESP32C3

Step 4 — Configure WiFi in Firmware

Open the provided .ino firmware file. At the top of the file, edit these constants to match the network the wristband will operate on:

```
#define WIFI_SSID "NetworkName"

#define WIFI_PASSWORD "NetworkPassword"

#define PC_IP "192.168.x.x" // IP of the PC running Phase Sync

#define HAPTIC_PIN_A 2 // GPIO D0 – Motor A

#define HAPTIC_PIN_B 10 // GPIO D10 – Motor B
```

Note: The WiFi network MUST be 2.4GHz. The XIAO ESP32C3 does not support 5GHz.

Step 5 — Flash

Connect XIAO to your computer via USB-C. Select the correct COM port under Tools > Port. Click Upload. Open Serial Monitor at 115200 baud and confirm you see: [Oneiro] WiFi connected

Step 6 — Test

With the circuit fully wired and firmware flashed, send a test UDP trigger to port 5001. Both motors should fire a 3-pulse pattern. Confirm Motor A and Motor B both vibrate independently when triggered by their respective GPIO pins.

Wristband Assembly

1. Solder JST PH 2.0 female connector to XIAO BAT+ and BAT- underside pads. Verify polarity with multimeter before connecting battery.
2. Breadboard test first — verify both motors fire on trigger before soldering anything permanently.
3. Transfer circuit to perfboard or direct solder once verified. Apply heat shrink to all exposed joints.
4. Mount XIAO and TP4056 on outer wrist side of elastic band.
5. Place battery flat along band between components.
6. Mount Motor A and Motor B on inner wrist, spaced apart, flat against skin with foam padding between motor and skin.
7. TP4056 USB port should remain accessible at band edge for recharging.
8. Full flex test — bend wrist in both directions, confirm no connections strain or break.
9. Charge LiPo fully before handing off.

Troubleshooting

Motors don't fire	Check 2N2222 pin orientation. Verify 220 Ohm on base. Confirm GPIO pins in
Motor runs constantly	Transistor wired backwards or base resistor missing. GPIO floating HIGH.
Weak vibration	Motor positive must connect to 5V/VIN not 3.3V.
Diode getting hot	Diode is reversed — cut power immediately, flip, re-test.
ESP32 won't connect WiFi	Must be 2.4GHz network. Check SSID/password. Re-flash if needed.

No Serial Monitor output	Wrong COM port selected. Try different USB-C cable (some are charge-only).
ESP32 resets constantly	Battery low or VIN/GND connection loose.

Questions during build? Contact the project owner before proceeding if anything is unclear. You're free to use AI if needed